

NGUYEN VIET KHANH

☎ Phone number: (+33) · 7 · 45 · 32 · 25 · 87, ✉ Email: nguyenvietkhanh.hn@gmail.com

🎓 Website: Research Gate, Google Scholar, vietkhanh-nguyen.github.io, 📍 Address: Evry, France

EDUCATION

Université Paris-Saclay

Université Évy Paris Saclay

2025 - Present

Master 2 - Smart Aerospace and Autonomous System

- **Scholarship:** IDEX scholarship

Hanoi University of Science and Technology (HUST)

School of Electrical Electronic Engineering

2021 - 2025

Bachelor - Control Engineering and Automation

- **Cumulative GPA:** 3.72/4.0, **Ranking:** 43/478,
- **Award:** Best thesis defend award, Student Research Conference (HUST) 2nd Place (2025) and 3rd Place (2024)
- **Scholarship:** VATM Excellent student scholarship

EXPERIENCE

Member of the Motion Control and Applied Robotics Lab

Supervised by Assoc.Prof.Dr Tung Lam Nguyen, and Dr. Thi Van Anh Nguyen

Jul 2022 - Aug 2025

Research Student

- **Research direction:** Anti-sloshing for liquid sloshing cylindrical containers; Simulation for robotic systems
- Developed control algorithms and validated models using **MATLAB/Simulink**.
- Programmed embedded systems in **C/C++** and performed physical simulation with **MuJoCo**.
- Built a solid background in control theory, motion control, and optimization.
- Experienced in scientific writing using **LaTeX** and presenting results at international conferences.

Member of the Computational Intelligence and Robotics Lab

Supervised by Prof. Chen-Chien Hsu

from Feb 2025 - Jun 2025

Internship Student

- **Research direction:** Multi-objective crowd-aware robot navigation using deep reinforcement learning
- Implemented learning-based algorithms and evolutionary computation techniques in **Python**.
- Simulated autonomous robots and tested navigation strategies in **ROS**.
- Gained adaptability and teamwork skills in an international research environment.

SKILLS AND RESEARCH INTEREST

Languages English (IELTS 7.0)

Simulation Simulink, Mujoco, ROS

Programming Matlab, Python, C/C++

Other tools: LaTeX, Git, Linux

Research interest: UAV, Control theory, Motion control, Predictive control, Optimization, Trajectory planning.

SELECTED PUBLICATIONS

- 1 **Khanh Nguyen Viet**, Hue Luu Thi, Thanh Cao Duc, Huy Nguyen Danh, Minh Nhat Vu, and Tung Lam Nguyen "Time-optimal motion planning and anti-sloshing control for a container under disturbances, IEEE Access (**Q1 Scopus**, doi.org/10.1109/ACCESS.2025.3533541)
- 2 **Khanh Nguyen Viet**, Minh Do Duc, Thanh Cao Duc and Lam Nguyen Tung "Anti-sloshing control: Flatness-based trajectory planning and tracking control with an integrated extended state observer", IET Cyber-Systems and Robotics (**Q2 Scopus**, doi.org/10.1049/csy2.12121)
- 3 **Viet Khanh Nguyen**, Hue Luu Thi, Duc Thanh Cao, Dang Huu Bang, and Tung Lam Nguyen "The Non-Flatness Property of the Liquid Sloshing System and an Approximate Approach" 2024 International Conference on Advanced Technologies for Communications Program (doi.org/10.1109/ATC63255.2024.10908250)
- 4 **Khanh Nguyen Viet**, Hue Luu Thi, Minh Do Duc, Thanh Cao Duc, Huy Nguyen Danh, and Nguyen Tung Lam "Input Shaping Integrated With Lyapunov Based Model Predictive Control For Anti-Sloshing Problem" 2024 3rd International Conference on Advances in Information and Communication Technology (**Q4 Scopus**, doi.org/10.1007/978-3-031-80943-9_94)